

Research Interests

Agentic Multimodal AI · Vision-Language Reasoning · Interactive and Verifiable AI Agents

Education

Pohang University of Science and Technology (POSTECH)

M.S. in Graduate School of Artificial Intelligence

Advisor: Prof. Jungseul Ok · GPA: 3.80/4.30

Pohang, South Korea

Mar 2025 – Feb 2027

Inha University

B.S. in Electrical Engineering and Artificial Intelligence (Double Major)

GPA: 3.97/4.50

Incheon, South Korea

Mar 2018 – Feb 2025

Publications

International

- ProCompNav: Proactive Instance Navigation with Comparative Judgment for Ambiguous User Queries**
Junhyuk Kwon, Seungjoon Lee, Hyejin Park, Kyle Min, and Jungseul Ok
Neural Information Processing Systems (NeurIPS), 2026 **(Oral)**
[arXiv](#) · [Code](#) · [Project Page](#)
- VIRO: Robust and Efficient Neuro-Symbolic Reasoning with Verification for Referring Expression Comprehension**
Hyejin Park, **Junhyuk Kwon**, Suha Kwak, and Jungseul Ok
Computer Vision and Pattern Recognition (CVPR), 2026
[Paper](#) · [arXiv](#) · [Code](#)
- MMTB: Evaluating Terminal Agents on Multimedia-File Tasks**
Chiyong Heo, Jaechang Kim, **Junhyuk Kwon**, Hoyoung Kim, Dongmin Park, Jonghyun Lee, and Jungseul Ok
Under review at International Conference on Learning Representations (ICLR), 2027
[arXiv](#) · [Code](#)
- SegmentWeave: Post-Processing Framework for Enhanced Dense Video Captioning**
Junhyuk Kwon (second author), et al.
Submitted to IEEE International Symposium on Multimedia (ISM), 2027 (Short Paper)
- RecKG: Knowledge Graph for Recommender Systems**
Junhyuk Kwon, Seokho Ahn, and Young-Duk Seo
ACM/SIGAPP Symposium on Applied Computing (SAC), 2024 **(Oral)**
[Paper](#) · [arXiv](#) · [Code](#)

Domestic

- Path-based Explainable Recommender System Leveraging Knowledge Graphs and Large Language Models**
Junhyuk Kwon, Seokho Ahn, and Young-Duk Seo
Korea Computer Congress (KCC), 2024 **(Undergraduate Paper Award)**

Work Experience

KT AI Research Intern, A-Intelligent Reasoning Team Researched and developed planning, evidence collection, and long-form report generation for Deep Research Agents.	Seoul, South Korea Jul – Aug 2026
Growdle AI Engineer Intern, AI Task Force Developed RAG-based document summarization and question-answering components for an LLM investment research agent.	Seoul, South Korea Jul – Sep 2024

Research Experience

Machine Learning Laboratory, POSTECH Graduate Researcher · Advisor: Prof. Jungseul Ok Researching agentic multimodal AI and vision-language reasoning through ProCompNav, VIRO, MMTB, and SegmentWeave.	Mar 2025 – Present
Knowledge-based Data Discovery Laboratory, Inha University Undergraduate Researcher · Advisor: Prof. Young-Duk Seo Conducted research on knowledge graphs, recommender systems, and large language models.	Mar 2023 – Feb 2025

Honors and Awards

Grand Prize (2nd Place), Inha AI Challenge — LLM-based Question Answering	Aug 2024
Undergraduate Paper Award, Korea Computer Congress	Jul 2024
Excellence Award (2nd Place), Hyundai Motor Group CTO AI Competition	Dec 2023

Additional Experience

NAVER Boostcamp AI Tech, Recommender Systems Track	Sep 2022 – Feb 2023
Korean Augmentation to the U.S. Army (KATUSA)	Jun 2019 – Jan 2021

Technical Skills

Python · PyTorch · C/C++ · Shell · SQL · Cypher · Git